

Task : File System Commands

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1. Objective -

The objective of this task was to perform basic file system commands on the target machine, locate and read files inside specific directories (loot)

2. Prerequisites – Linux Permissions Model

Every file and directory on a Linux system has an owner, a group, and a set of permissions that control who can **read (r)**, **write (w)**, or **execute (x)** it.

Permissions are split into three categories:

- **User (u)** — the owner of the file
- **Group (g)** — users belonging to the file's group
- **Others (o)** — everyone else on the system

When listed with `ls -la`, permissions appear as `-rw-r--r--`. The first character indicates file type (- for a regular file, d for a directory) and the remaining characters are three sets of `rwX` for user, group, and others respectively.

3. Step-by-step process

Step 1: Connecting and Confirming Identity

After using SSH to connect to the target machine, I confirmed my session context:

```
student@hacking:~$ pwd
/home/student
student@hacking:~$ whoami
student
```

This confirmed I was logged in as the `student` user.

Step 2: Checking the home directory

```
student@hacking:~$ ls
loot  practice  read_secret.sh  run_secret
secret.txt
```

This showed two directories (`loot`, `practice`) and three files, including a script (`read_secret.sh`), a binary (`run_secret`), and a plain file (`secret.txt`)

Step 3: Exploring the `loot` Directory

Basic navigation (`cd`) and file reading (`cat`) confirmed access to this directory's contents

```
student@hacking:~$ cd loot
student@hacking:~/loot$ ls
readme.txt
student@hacking:~/loot$ cat readme.txt
flag{you_know_cd_ls_cat_now}
```

Digging deeper with `ls -la` showed the hidden files, denoted by a leading dot, along with full permission details :

```
student@hacking:~/loot$ ls -la
total 16
drwxrwxr-x 2 student student 4096 Jan 28 06:05 .
drwxr-x--- 6 student perm   4096 Apr 13 11:10 ..
-rw-rw-r-- 1 student student   32 Jan 28
06:01 .hidden_file.txt
-rw-r--r-- 1 student student   29 Jan 28 06:00 readme.txt
```

What `ls -la` does: By default, `ls` hides any file or directory whose name begins with a dot (`.`), it is the Linux convention for "hidden" files. The `-a` flag ("all") tells `ls` to include these. The `-l` flag switches to "long format," showing permissions, ownership, size, and modification date for each entry.

This revealed a hidden file, `.hidden_file.txt`, which does not appear under a plain `ls`. Since it was owned by `student` with read permission granted to the owner (`rw-`), I was able to read it directly:

```
student@hacking:~/loot$ cat .hidden_file.txt
flag{you_found_the_hidden_file}
```

What cat does: cat (concatenate) prints the full contents of a file directly to the terminal — the standard way to quickly view a small text file without opening an editor.

Step 5: Screenshot Evidence

```
student@hacking:~$ ls
loot  practice  read_secret.sh  secret.txt
student@hacking:~$ cd loot
student@hacking:~/loot$ ls
readme.txt
student@hacking:~/loot$ cat readme.txt
flag{you_know_cd_ls_cat_now}
student@hacking:~/loot$ ls -la
total 16
drwxr-xr-x 2 student student 4096 Jan 28 06:05 .
drwxr-xr-x 6 student perm  4096 Apr 13 11:10 ..
-rw-rw-r-- 1 student student 32 Jan 28 06:01 .hidden_file.txt
-rw-rw-r-- 1 student student 29 Jan 28 06:00 readme.txt
student@hacking:~/loot$ cat .hidden_file.txt
flag{you_found_the_hidden_file}
student@hacking:~/loot$ pwd
/home/student/loot
student@hacking:~/loot$ cd ..
student@hacking:~$ cd practice
student@hacking:~/practice$ ls
```

What is the name of the file located inside the "loot" folder?

readme.txt

✓ Correct Answer

What are the contents of the file which was in the "loot" folder?

flag{you_know_cd_ls_cat_now}

✓ Correct Answer

What are the contents of the hidden file in the "loot" folder?

flag{you_found_the_hidden_file}

✓ Correct Answer

4. Conclusion

This task built on the earlier SSH login exercise by exploring Linux file navigation (`cd`, `ls`), file viewing (`cat`), and viewing hidden files using (`ls -la`), this shows that hidden doesn't mean security as the hidden files could be accessed.